Started on	Thursday, 15 October 2026, 10:11 AM
State	Finished
Completed on	Thursday, 15 October 2026, 10:54 AM
Time taken	43 mins 15 secs

Question 1 | Correct Marked out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Input:

A positive Integer n

Output:

Print the value of the counter variable.

For example:

Input	Result
12	11

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    int count=0;
    scanf("%d",&n);
    count++;
    int rev=0,remainder;
    count++;
    while(n!=0){
        remainder=n%10;
        rev=rev*10+remainder;
        n/=10;
        count+=4;

    }
    count++;
    printf("%d",count);
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Question 2 | Correct Marked out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as “No”.

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value “x”)

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int pair(int arr[],int left,int right,int x) {
    if (left>=right)
        return 0;
    int sum=arr[left]+arr[right];
    if (sum==x){
        printf("%d\n%d\n",arr[left],arr[right]);
        return 1;
    }
    if (sum<x)
        return pair(arr,left+1,right,x);
    else
        return pair(arr,left,right-1,x);
}
int main() {
    int n,x;
    scanf("%d",&n);
    int arr[n];
    for (int i=0;i<n;i++)
        scanf("%d",&arr[i]);
    scanf("%d",&x);
    if (!pair(arr,0,n - 1,x))
        printf("No\n");
    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔

	Input	Expected	Got	
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔